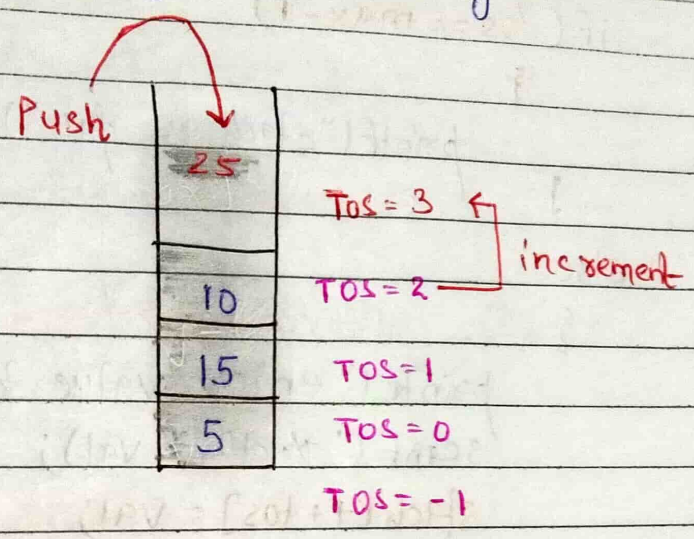
→ **Push** → Push is an operation that inserts an elements onto the top of the stack, increasing its size by one.



Algorithm:-

Step 1: IF TOS == max-1
print" stack is full"

Step 2: else,
stack[++tos] = val

Step 3: Exit ()

Akash

→ Pseudo code for push in stack :-

```
int val;
if( tos == max-1)
    {
        printf("stack is full");
    }
else
    {
        printf("enter value to insert\n");
        scanf("%d", &val);
        stack[++tos] = val;
    }
```

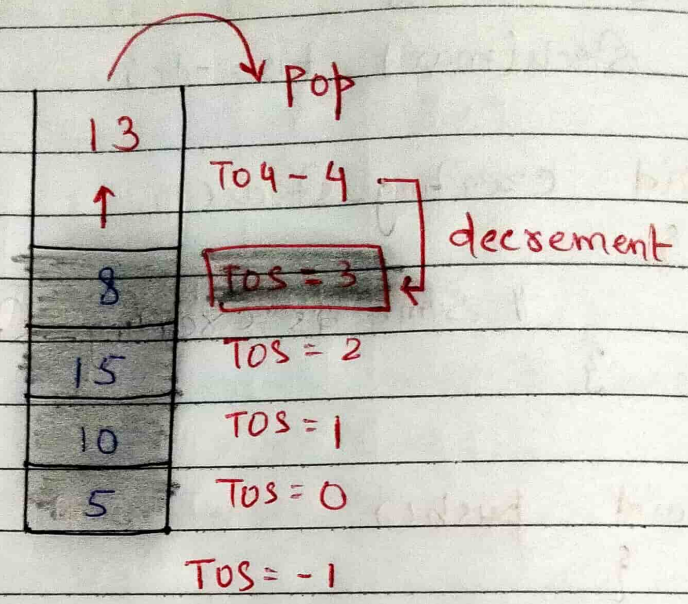
Akash

→ // C program for push in stack;

```
#include<stdio.h>
#define max 100
int stack[max], tos=-1;

void creating_stack()
{
        // Same as creating_stack
}

void push()
{
        // same as pseudo code for push in stack
}

void display()
{
        // same as display
}

int main()
{
        creating_stack();
        push();
        display();
        return 0;
}
```

Akash

→ **Pop :-** Pop is an operation that removes the top element from the stack, decreasing its size by one.



Algorithm :-

Step 1: IF tos == -1
print " empty stack"

Step 2: else,
stack[--tos]

Step 3: exit

→ Pseudo code for pop in the stack:

```
if (tos==-1)
 {
        printf("empty stack");
 }
else
 {
        stack[--tos];
 }
```

Akash

→ // : c program for pop.

```
#include<stdio.h>
#define max 100
int stack[max], tos=-1;

void creating_stack()
{
    // same as ~~stack~~ creating_stack
}

void pop()
{
    // same as pseudo code for pop
}

void display()
{
    // same as display
}

int main()
{
creating_stack();
pop();
display();
return 0;
}
```

Akash

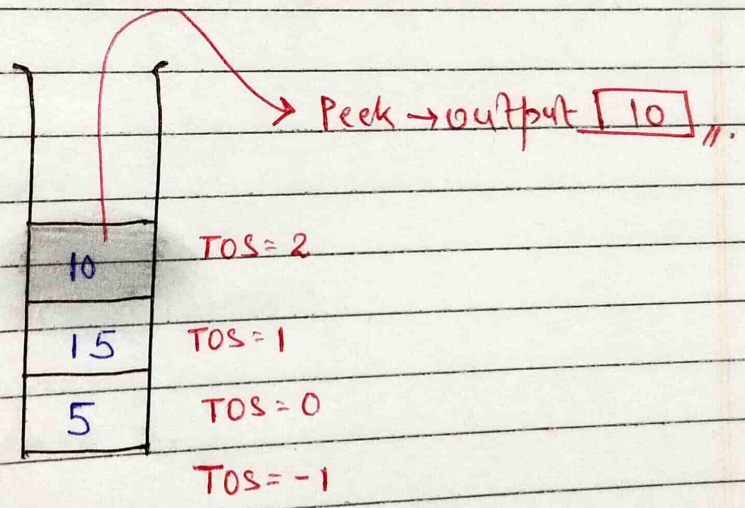
→ **peek :-** Peek is an operation that returns the top elements of the stack without removing it.

properties of peek :

(i) Does not modify the stack.

(ii) Return the last inserted element (top of stack).

(iii) If the stack is empty, it should handle the condition properly.



Algorithm :

Step 1: IF (tos == -1)
print "empty stack"
else,
Step 2: print " stack [tos]

Step 3: exit

Akash

→ pseudo code for peek :

```
if ( tos == -1)
  {
      printf(" empty stack");
  }
else
  {
      printf(" %d", stack[tos]);
  }
```

Akash

→ // C program for peek;

```
#include<stdio.h>
#define max 100
int stack[max], tos=-1;

void creating_stack ()
{
        // same as creating_stack
}

void peek()
{
        // same as pseudo code for peek
}

int main()
{
        creating_stack();
        peek();
        return 0;
}
```

Akash